

Attribute Description

Flight Info

- `flight_id`: unique ID
- `aircraft_id`: aircraft tail number (categorical, ~200 planes)
- `flight_hours`: cumulative hours since last maintenance (0–600 hrs, typical A-check cycle ~500–800 hrs)
- `flight_cycles`: takeoff-landing cycles since last maintenance (0–300, cycles wear parts faster than hours)

Sensor data (engine + parts)

- `engine_temp`: °C (normal 200–900°C, spikes above 950°C → risk)
- `vibration_level`: g-force (0.1–5 g, >3 g = unusual stress)
- `oil_pressure`: psi (30–80 normal, <25 = risk)
- `fuel_flow`: kg/hr (2000–8000, inefficiency spikes before failure)
- `cabin_pressure`: psi (10–15 normal, <8 = problem)

Environmental/Route factors

- `route_type`: Domestic / International / Long-haul (categorical)
- `avg_weather_temp`: °C (–30 to 45)
- `turbulence`: scale 0–5 (higher = stress)
- `airport_condition`: 0–1 (bad runway, dust, heat, snow, etc.)

Maintenance History

- `last_maintenance_days`: days since last major check (1–180)
- `maintenance_score`: rating of last check (0.8–1.0 = good, 0.4–0.7 = rushed/poor)

Target Variable

- `failure_within_30_days`: binary (1 = failed within 30 days after this flight, ~2–5% prevalence overall, higher under extreme stress conditions).